VERY EASY & CUTE SECTION
WITH MANY MINI ANSWERS

VSAQ
SECTION-B

Baby Bullet Vs
Baby Boss

Minimum Reading
Maximum Scoring

VSAQ CHAPTERS

28 MARKS

★★½	1.	ELECTRIC CHARGES & FIELDS	(2 Marks)	62
★★	2.	ELECTROSTATIC POTENTIAL	(2 Marks)	63-64
★★½	3.	CURRENT ELECTRICITY	(2 Marks)	65-66
★★★	4.	MOVING CHARGES & MAGNETISM	(2 Marks)	67
★★★	5.	MAGNETISM AND MATTER	(2 Marks)	68
★★	6.	ELECTROMAGNETIC INDUCTION	(2 Marks)	69-70
★★★	7.	ALTERNATING CURRENT	(2 Marks)	71 - 72
★★	8.	ELECTRO MAGNETIC WAVES	(2 Marks)	73
★★½	9.	RAY OPTICS	(2 Marks)	74
★★½	10.	WAVE OPTICS	(2 Marks)	75
★★★	11.	DUAL NATURE	(2 Marks)	76
★★½	12.	ATOMS	(2 Marks)	77
★★★	13.	NUCLEI	(2 Marks)	78
★★★	14.	SEMICONDUCTOR ELECTRONICS	(2 Marks)	79-80

Super Scoring Pages

★★★SSP★★★

Pages 62, 63, 65, 67, 68, 69, 71, 73, 74, 75, 76, 77, 78, 79

Dear Dude, Please note that VSAQs in each chapter are jumbled, and Easy VSAQ's are brought to the beginning and some VSAQs are pushed back to **STAR Questions plus.** To gain **maximum score**, read all the Qs given in the STAR Questions plus.

1 ELECTRIC CHARGES & FIELDS

2 Marks

78. The electric lines of force do not intersect. Why ? **[BMP]**

A : If electric lines of force intersect, the electric field at the point of intersection will have two directions simultaneously, which is physically impossible. So, the electric lines of force do not intersect.

---

79. What is meant by the statement charge is quantised ?

A : **Quantisation of charge:** The electric charge(q) is always equal to an integral multiple of the electron charge (e). Thus $q = ne$. Here $n = 0, \pm1, \pm2, \pm3, \ldots$ and $e = 1.6\times10^{-19}$C.

---

80. How many electrons constitute 1C of charge ?

A: Given that $q = 1C$, $e = 1.6 \times 10^{-19}$ C, $n = ?$

Formula: $q = ne \Rightarrow n = \frac{q}{e} = \frac{1}{1.6\times10^{-19}} = 0.625 \times 10^{19}$ electrons

---

81. What is the importance of Gauss law in electrostatics.

A : **Importance of Gauss's Law:**

1) Gauss's law is **true for any closed surface** (of any shape and any size).
2) It gives the relation between the **electric field and the charge.**
3) It is valid for **stationary charges** as well as for **rapidly moving charges.**
4) Gauss's law is based on Coulomb's law. So Gauss' law is valid as long as Coulomb's law is valid.

---

82. Write the expression for electric intensity due to an infinitely long straight uniformly charged wire, at a radial distance r from the wire. Explain the terms in it.

A : Electric intensity due to an infinite long charged wire is $E = \frac{\lambda}{2\pi\varepsilon_0 r}$

Here λ is uniform linear charge density of the wire, r is radial distance from the wire and ε_0 is permittivity of free space.

---

83. Write the expression for electric intensity due to an infinite uniformly charged plane sheet. Explain the terms in it.

A: Electric intensity due to an infinite plane sheet of charge is $E = \frac{\sigma}{2\varepsilon_0}$

Here σ is uniform surface charge density of the sheet and ε_0 is permittivity of free space.

[☞Remaining VSAQs are in Page 91]

2 ELECTROSTATIC POTENTIAL

2 Marks

84. Can there be electric potential at a point with zero electric intensity? Give an example.

A : Yes.

Ex: Inside a charged metal spherical shell, potential is non-zero but electric intensity is zero .

---

85. Can there be electric intensity at a point with zero electric potential ? Give an example.

A : Yes.

Ex: When two charges +q and –q are separated by a distance, electric intensity at the mid point is non-zero. But the potential at the midpoint is zero [∵ + V–V = 0.]

---

86. The potential difference between two points is 1 V.

What is the work done to move 1 C charge between them?

A : Given potential difference V =1Volt and charge q=1C

∴Work done W=Vq =1×1= 1 Joule

---

87. Why is the electric field always at right angles to the equipotential surface ? Explain.

A : The potential difference between any two points on an equipotential surface dV=0.

Potential difference $dV = -\vec{E}.d\vec{r} \Rightarrow 0 = -Edr\cos\theta \Rightarrow \cos\theta = 0 \Rightarrow \theta = 90°$

∴Electric field $\vec{E}$ is always at right angles to the equipotential surface.

---

88. Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel.

a) What is the ratio of charges on each capacitor?

b) What is the ratio of potential differences across each capacitor?

A : Given that $C_1 = 1\mu F$, $C_2 = 2\mu F$, $C_3 = 3\mu F$

In the parallel combination of capacitors we have $Q \propto C$ and V is same,

a) $Q_1 : Q_2 : Q_3 = C_1 : C_2 : C_3 = 1 : 2 : 3$

b) $V_1 = V_2 = V_3 = 1 : 1 : 1$ [∵ V is same]

89. Three capacitors of capacitances 1 μF, 2μF and 3 μF are connected in series. a) What is the ratio of charges? b)What is the ratio of potential differences?

A: Given that $C_1 = 1\mu F, C_2 = 2\mu F, C_3 = 3\mu F$

In the series combination of capacitors we have Q is same and $V \propto \frac{1}{C}$

a) $Q_1 : Q_2 : Q_3 = 1 : 1 : 1$ [$\because$ Q is same]

b.) $V_1 : V_2 : V_3 = \frac{1}{C_1} : \frac{1}{C_2} : \frac{1}{C_3} = \frac{1}{1} : \frac{1}{2} : \frac{1}{3} = \frac{6}{1} : \frac{6}{2} : \frac{6}{3} = 6 : 3 : 2$

---

90. What happens to the capacitance of a parallel plate capacitor if the area of its plates is doubled?

A: Capacitance $C = \frac{\varepsilon_0 A}{d} \Rightarrow C \propto A$ Here $A_2 = 2A_1$

$\therefore \frac{C_2}{C_1} = \frac{A_2}{A_1} = \frac{2A_1}{A_1} = 2 \Rightarrow C_2 = 2C_1$ ∴Capacitance becomes **double.**

---

91. What happens to the capacitance of a parallel plate capacitor if the distance between plates is doubled?

A: Capacitance $C = \frac{\varepsilon_0 A}{d} \Rightarrow C \propto \frac{1}{d}$ Here $d_2 = 2d_1$

$\therefore \frac{C_2}{C_1} = \frac{d_1}{d_2} = \frac{d_1}{2d_1} = \frac{1}{2} \Rightarrow \frac{C_2}{C_1} = \frac{1}{2} \Rightarrow C_2 = \frac{C_1}{2}$ ∴Capacitance becomes **half.**

---

92. In a hydrogen atom, the electron and proton are at a distance of 1Å . [BMP] Find the dipole moment of the system.

A: Given length of dipole is $2a = 1Å = 1 \times 10^{-10}$ m, we know $q = 1.6 \times 10^{-19}$ C, p = ?

Formula: Dipole moment $p = q(2a) = 1.6 \times 10^{-19} (1 \times 10^{-10}) = \mathbf{1.6 \times 10^{-29}}$ **C-m**

---

93. What is electrostatic shielding? Give an example.

A : **Electrostatic shielding** is the process of isolating a specific region from external electric fields by enclosing it within a conducting material.

Ex: Metallic car body protects passengers from thunderstorm as the electric field inside is zero.



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[☞Remaining VSAQs are in Page 92]

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3 CURRENT ELECTRICITY

2 Marks

94. State Ohm's Law and write its mathematical form.

A : **1) Ohm's Law :** At constant temperature, the potential difference (V) between the ends of a conductor is directly proportional to the current (I) passing through it.



2) $V \propto I \Rightarrow V = RI$ (or) $I = \frac{V}{R}$ (or) $R = \frac{V}{I}$

Here R is the resistance of the conductor.

---

95. Write the vector form of Ohms law. Explain the terms in it. [BMP]

A : **Vector form of ohm's law:** Current density $\vec{j} = \sigma \vec{E}$.

Here σ is conductivity and $\vec{E}$ is the electric field intensity.

---

96. State Kirchhoff rules for electric network.

A: **1) Kirchoff's First Law :** The algebraic sum of currents meeting at any junction in an electric circuit is zero. Total current enters the junction is equal to total current leaving.

2) Kirchoff's Second Law : The algebraic sum of changes in potential around any closed loop involving resistors and cells is zero.

---

97. Define temperature coefficient of resistance.

A: The **temperature coefficient of resistance**(α) is the ratio of change in **resistance per unit** original resistance per degree centigrade change in temperature. $\left[\because \alpha = \frac{\Delta R}{R_0 \Delta T}\right]$

---

98. Define resistivity or specific resistance.

A: **1) Resistivity** (ρ) is the resistance of a conductor per unit length per unit area of cross section .

2) Formula: $\rho = \frac{RA}{l}$. Its SI unit is Ωm.

99. Two wires of equal length, of copper and manganin, have the same resistance. Which wire is thicker? (Given $\rho_{manganin} > \rho_{copper}$)

A : We know that resistivity $\rho = \frac{RA}{l}$. As length l and resistance R are constant, $\rho \propto A$

Resistivity ρ is high for **manganin.**

Hence **manganin has more area and more thicker.**

---

100. If the voltage V applied across a conductor is increased to 2V, how will the drift velocity of the electrons change?

A: Drift velocity $v_d = \frac{I}{neA} = \frac{V}{neAR}$ $\left(\because I = \frac{V}{R}\right) \Rightarrow v_d \propto V$ ($\because$ n,e,A,R are constants)

$\therefore \frac{v_d^1}{v_d} = \frac{V^1}{V} = \frac{2V}{V} = 2$ $\qquad \therefore v_d^1 = 2v_d$.

Thus, the **drift velocity becomes double.**

---

101. Define mobility of charge carriers.

A: **1) Mobility(μ)of a charge carrier** is the magnitude of its drift velocity (v_d) per unit electric field strength (E).

2) Formula: $\mu = \frac{v_d}{E}$

---

102. Electrical power is transmitted from power stations to homes and factories at high voltages, why?

A : **1) Electric power** is transmitted at **high voltages** in primarily to reduce heat energy loss.

2) Power P= VI $\Rightarrow$ V and I are related inversely and also Heat loss = I^2R

$\therefore$ High voltages give low current and hence low heat energy loss.

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[☞ Remaining VSAQs are in Page 92]

4 MOVING CHARGES & MAGNETISM

2 Marks

103. What is the importance of Oersted's experiment?

A : Importance of Oersted's Experiment :

Oersted's experiment established the connection between electric current and magnetism. It showed that moving charge (current) produces magnetic field in its surrounding space.

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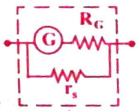
104. What is the principle of moving coil galvanometer?

A: Principle of MCG: The current carrying coil placed in a uniform magnetic field experiences a **torque.**

---

105. How do you convert a moving coil galvanometer into an ammeter?

A: A galvanometer (G) can be converted into an ammeter (A) by connecting a very **small shunt resistance** (r_s) in parallel with it.

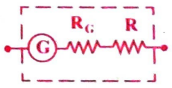


A

---

106. How do you convert a moving coil galvanometer into a voltmeter?

A : A galvanometer (G) can be converted into a voltmeter (V) by connecting a **high resistance (R)** in series with it.



V

---

107. Write the formula for the magnetic field due to infinite straight wire and explain the terms in it. **[BMP]**

A : Magnetic field due to infinite straight wire is $B = \frac{\mu_0 I}{2\pi r}$. Here μ_0 is the permeability of free space, I is electric current, r is the perpendicular distance from line.

---

108. Two parallel conducting wires, separated by a distance d carry I_1 and I_2 currents. When do they attract and repel? What is the magnitude of force per unit length acting on any wire?

A : 1) The two parallel current conducting wires attract when the currents flow in the same direction. They repel when the currents flow in the opposite direction.

2) The magnitude of the force per unit length is $f = \frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$

Here μ_0 is permeability of free space and d is the distance between the wires.

[☞ Remaining VSAQs are in Page 93]

5 MAGNETISM & MATTER

2 Marks

109. Magnetic lines of force don't intersect. Why?

A: Any two magnetic field lines do not intersect each other because if they do, it means that at the point of intersection it shows two different directions which is impossible.

---

110. Magnetic lines form continuous closed loops. Why?

A : The magnetic field lines of a magnet or a solenoid form continuous closed loops because the magnetic poles N and S always exist together in pairs.

---

111. Define magnetisation of a material and write its formula.

A: **Magnetisation** (M) is defined as the **net magnetic dipole moment per unit volume** of the material. **Formula:** $M = \frac{m_{net}}{V}$

---

112. What is the magnetic moment associated with a solenoid?

A : **Magnetic moment** of a solenoid is $\bar{M} = NI\bar{A}$. Here N is the number of turns, I is the current passing through it and $\bar{A}$ is the area of cross section.

---

113. Define magnetic susceptibility and relative permeability. **[BMP]**

A: 1) **Magnetic Susceptibility** (χ) is the ratio between **magnetisation (M)** and **magnetic intensity (H) Formula:** $\chi = \frac{M}{H}$. It has no units. It determines how easily a material can be magnetized.

2) **Relative permeability**(μ_r) is the ratio of the magnetic permeability of a specific medium (μ) to the magnetic permeability of free space (μ_0) $\mu_r = \mu/\mu_0$

---

114. State Gauss law in magnetism and write it its mathematical expression.

A: 1) **Gauss law:** The net magnetic flux (ϕ) through any closed surface is equal to zero.

2) **Mathematical expression:** $\oint \bar{B}.d\bar{S} = 0$

Here $\bar{B}$ is the magnetic vector, and $d\bar{S}$ is surface vector area.

---

115. Classify the following materials with regard to magnetism.
Iron, Platinum, Copper, Magnesium.

A: 1) **Ferromagnetic material:** Iron

2) **Paramagnetic material:** Platinum, Magnesium

3) **Diamagnetic material:** Copper

6 ELECTRO MAGNETIC INDUCTION

2 Marks

116. State Faraday's law of electromagnetic induction. **[BMP]**

A : 1) **Faraday's law:** The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

2) **Mathematical expression:** Induced emf $\varepsilon = -\frac{d\phi_B}{dt}$

In the case of a coil of N turns, $\varepsilon = -N\frac{d\phi_B}{dt}$

---

117. State Lenz's law.

A : 1) **Lenz's Law (Neumann's law) :** The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux.

2) Faraday's law of induction: $\varepsilon = -\frac{d\phi_B}{dt}$. The negative sign indicates Lenz's law.

---

118. Define magnetic flux.

A : 1) **Magnetic Flux** is defined as the total number of magnetic field lines passing normally through the given plane.

2) **Formula:** $\phi_B = \bar{B}.\bar{A} = BA\cos\theta$. Here θ is the angle between $\bar{A}$ and $\bar{B}$.

Magnetic flux is a scalar. Its SI unit is Tm^2

---

119. What is the working principle of A.C. generator?

A: AC generator works on the principle of **electromagnetic induction.**

(The AC generator converts mechanical energy into electrical energy by rotating a coil with in a magnetic field, which changes the magnetic flux and induces an alternating emf.)

---

120. Write the expression for self-induced emf and explain the terms therein.

A: The expression for self induced emf is $\varepsilon = -L\frac{dI}{dt}$

Here L is the self inductance, $\frac{dI}{dt}$ is the rate of change of current.

121. What is self-induction?

A: 1) **Self induction** is the magnetic flux linked with the coil when unit current flows in it. (or) When the current in a coil changes, an emf is induced in the coil opposing the change in the current through the coil. This phenomenon is called self induction.

2) $\therefore \phi \propto I \Rightarrow \phi = LI$ (or) $Emf = -L\frac{dI}{dt}$

---

122. What is mutual induction?

A: **1) Mutual induction** is defined as the flux linked with secondary coil when unit current flows through the primary coil. (or) When the current in one coil changes, an emf is induced in another coil kept near it. This phenomenon is called mutual inductance.

2) $\phi \propto I \Rightarrow \phi = MI$ (or) $Emf = -M\frac{dI}{dt}$

---

123. Why do birds fly off a high tension wire when the current is switched 'ON'?

A: 1) When current is switched on, **induced current flows** through the body of the bird.

This induced current flows in the opposite direction through the wings of the bird.

2) Hence, the wings experience a force of mutual repulsion.

Thus, the wings spread and hence the birds fly due to electromagnetic induction.

---

124. Why is spark produced in the switch of a fan when it is switched off?

A: A fan motor contains coils of high inductance. When the fan is switched off, the current drops to zero rapidly. This sudden change induces a very high back emf (voltage) across the switch contacts. This high voltage is strong enough to ionize the air in the small gap of the switch, resulting in a **visible spark.**



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[☞ Remaining VSAQs are in Page 94]

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7 ALTERNATING CURRENT

2 Marks

125. What is transformer ratio?

A : **1) Transformer Ratio** is the ratio of output emf to input emf of a transformer. (or) It is the ratio between number of turns in the secondary coil and the number of turns in the primary coil of a transformer.

2) Formula: Transformer ratio $= \frac{N_s}{N_p} = \frac{V_s}{V_p}$

---

126. A transformer converts 200 V AC into 2000 V AC. Calculate the number of turns in the secondary if the primary has 10 turns.

A : Given $V_p = 200$ V, $V_s = 2000$ V, $N_p = 10$, $N_s = ?$

Transformer ratio formula : $\frac{N_s}{N_p} = \frac{V_s}{V_p} \Rightarrow N_s = \frac{V_s}{V_p} N_p = \frac{2000}{200} \times 10 = 100$ turns

---

127. Write the expression for the reactance of (i) an inductor and (ii) a capacitor.

A : **(i) Reactance of an inductor** is $X_L = \omega L$
Here ω is angular frequency and L= Inductance

(ii) Reactance of a capacitor is $X_C = \frac{1}{\omega C}$
Here ω is angular frequency and C = Capacitance

---

128. A Capacitor has a capacitance of $\frac{1}{\pi}\mu F$. Find its reactance for a frequency of 50 Hz.

Sol: Given Frequency t = 50Hz , Capacitance $C = 1\times10^{-6}$ F **[BMP]**

Reactance $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC} = \frac{1}{2\pi\times50\times\frac{1}{\pi}\times10^{-6}} = \frac{1}{100\times10^{-6}} = \frac{1}{10^{-4}} = 10k\Omega$

---

129. A pure inductor of 21mH is connected to a source of 220V. Find the inductive reactance if the frequency of the source is 50Hz.

A: Given Inductance $L = 21mH = 21\times10^{-3}$ H, Voltage V = 220V, Frequency f = 50Hz

$\therefore$ Inductive reactance $X_L = 2\pi fL = 2x\ 3.14\times50\times21\times10^{-3} = 6.597\Omega = 6.6\Omega$

130. What is the phase difference between A.C emf and current in the following: pure resistor, pure inductor and pure capacitor.

A : 1) In a **pure resistor**, the phase difference between AC voltage and AC current is **zero.**

2) In a **pure Inductor**, the **current lags** behind the voltage by $\pi/2$.

3) In a **pure capacitor**, the **current leads** ahead the voltage by $\pi/2$.

---

131. Define power factor. On which factors does the power factor depend?

A : 1) **Power Factor** is the cosine of the phase angle (ϕ) between the voltage and the current.

Formula 1: $P = V I \cos\phi$. Here $\cos\phi$ is called power factor.

Formula 2: Power factor $\cos\phi = \frac{R\,(\text{Resistance})}{Z\,(\text{Impedance})}$

2) Power factor depends on the phase difference between voltage and current or equivalently on the ratio R/Z.

---

132. What is meant by wattless component of current?

A : **Wattless current :** In a purely inductive or capacitive circuit, the power factor $\cos\phi=0$. Then no power is dissipated even though current is flowing in the circuit.

In such cases, the current is said to be **wattless current.**

---

133. Why capacitor blocks Direct Current (D.C.)?

A: D.C. has constant magnitude and direction. Once a capacitor is fully charged, it behaves like an open circuit for D.C. When current passes through the capacitor it blocks direct current D.C.

---

134. On what principle metal detector works? Explain it.

A: 1) Metal detector works on the principle of **electromagnetic induction.**

2) The primary search coil in metal detector generates an alternating magnetic field. When a metal object is nearby, due to the influence of the magnetic field, current will be induced in the metal. The induced current creates a secondary magnetic field which is detected by the metal detector's receiver coil, and it generates a sound as an alarm.

---

135. How the phenomenon of resonance in the series LCR circuit be used in tuning mechanism of a radio and TV set?

A: In the series LCR circuit, energy alternates between the inductor and capacitor, enabling the radio or TV to pick up a precise frequency. **Resonance in the circuit** enables radio or T.V. tuning by selecting a specific station's frequency while rejecting others.

8 ELECTRO MAGNETIC WAVES

2 Marks

136. Give two uses of infrared rays. **[BMP][AP 18,19,22,23]**

A : 1) Infrared rays are used in **remote control systems of TV**

2) Infrared rays are used for the **treatment of skin diseases.**

3) Infrared rays are used in **earth satellites.**

4) Infrared rays are used for taking photographs where there is no visible light due to fog or mist.

---

137. What are the applications of microwaves? **[AP 15,16,17,18,19,25]**

A: **Applications of microwaves:**

1) Microwaves are used in **electrical ovens** for cooking and heating.

2) Microwaves are used in **wi-fi network**, satellite communication.

3) Microwaves are used for deep **muscle pain relief.**

4) Microwaves are used in **radar systems** due to their short wavelengths .

---

138. What is the principle of production of electromagnetic waves? **[AP 19, 24]**

A: 1) Electromagnetic waves are produced by **accelerating or oscillating charged particles.**

2) When a charged particle like an electron vibrates or changes speed, it creates oscillating electric and magnetic fields that ripple outward together as self-propagative EM waves.

---

139. What is the displacement current, and what is its formula?

A: 1) Displacement current (I_d) is equal to ε_0 times the rate of change of electric flux through the surface.

2) Formula: $I_d = \varepsilon_0 \frac{d\phi}{dt}$ where $\frac{d\phi}{dt}$ is the rate of change of electric flux.

The displacement current occurs when there is change in electric flux.

---

140. The charging current for a capacitor is 0.6 A. **[AP 20]**
What is the displacement current across its plates?

A : The displacement current is 0.6 A.
Reason: The displacement current across the plates of a capacitor is always equal to charging current of the capacitor.

---

141. What is the wavelength range of X-rays?

A: The wavelength range of X-rays: **10^{-3} nm to 10 nm** (or) **0.01 Å to 100 Å**

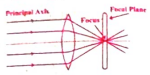
[☞ Remaining VSAQs are in Page 94]

9 RAY OPTICS

2 Marks

142. What do you understand by the focal plane? Explain with a neat ray diagram.

A: The light rays coming from an object at infinity, when pass through a lens converge (or appear to diverge) at a **point** to form a clear image. The plane passing through **that point and perpendicular** to the principal axis of the lens is called **focal plane**.



---

143. Define power of a convex lens. What is its unit?

A: **1) Power of a convex Lens** is defined as the reciprocal of its focal length (f).

Power $P = \frac{1}{f}$ **SI unit :** dioptre (D)

2) Power of a lens is its ability to converge incoming light rays.

---

144. Two thin lens of power +4D and -2D are in contact.
What is the focal length of the combination?

A: Given $P_1 = +4D$, $P_2 = -2D$; Equivalent power $P_{eq} = P_1 + P_2 = 4 + (-2) = +2D$

Focal length $f = \frac{1}{P_{eq}} = \frac{1}{2} = 0.5m = 50\ cm$

---

145. What is the critical angle? Give one application of total internal reflection? [BMP]

A: **1) Critical Angle (i_C):** When a light ray travels from denser medium to rarer medium, the angle of incidence for which the angle of refraction 90° is called critical angle.

2) Applications of TIR : Optical fibre, Mirage.

---

146. State the conditions under which total internal reflection takes place.

A: **Conditions for TIR:**

(i) Light must travel from a denser to rarer medium.

(ii) The angle of incidence at the interface must be greater than the critical angle. ($i > i_C$)

---

147. A small angled prism of 4° deviates a ray through 2.4°.
Find the refractive index of the prism.

A: Given angle of prism A = 4°, angle of deviation $D_m = 2.4°$, refractive index $n_{21} = ?$

Formula: For a thin angled prism $D_m = (n_{21} - 1)\ A$

$\Rightarrow 2.4 = (n_{21} - 1)\ 4 \Rightarrow n_{21} - 1 = \frac{2.4}{4} = 0.6 \Rightarrow n_{21} = 1 + 0.6 = 1.6$

[☞ Remaining VSAQs are in Page 95]

10 WAVE OPTICS

2 Marks

148. State Malus' law. Write the expression of Malus' law. [BMP]

A : **1) Malus' law:** The intensity of plane polarized light transmitted through an analyzer varies as the square of cosine of angle between transmission axes.

2) Expression: $I = I_0 \cos^2\theta$

Here I_0= initial intensity, I = transmitted intensity, θ= angle between axes

---

149. What is polarization of light? Which field vector, electric or magnetic, is used to represent the polarization of an electromagnetic waves?

A: **1) Polarization of light** is the phenomenon of restricting vibrations of light to one direction perpendicular to propagation.

2) The electric field vector $\vec{E}$ is used to represents the polarization of an electromagnetic wave.

---

150. Two identical coherent waves, each o f intensity I_0, are producing an interference pattern. Write the value of resultant intensity at a point of
(1) constructive interference (2) destructive interference.

A: 1) For constructive interference, (when $I_1 = I_2 = I_0$), $I_{max} = \left(\sqrt{I_0} + \sqrt{I_0}\right)^2 = 4I_0$

2) For destructive interferences, $I_{max} = \left(\sqrt{I_0} - \sqrt{I_0}\right)^2 = 0$

---

151. If a wave undergoes reflection and refraction, what will be the phase change in each case?

A: 1) In **reflection** if light reflects from a denser medium there is a phase change of **π radians.**
2) In **refraction**, there is a **no phase change.**

---

152. Compare Maxwell's Electromagnetic theory with Huygen's theory of light?

A:

Maxwell Electromagnetic Theory	Huygen's theory of light
1) Light is an **electromagnetic wave.**	1) Light is an transverse **mechanical wave.**
2) Explains electric and magnetic nature	2) Explains reflection and refraction
3) Explains polarization	3) Does not explain polarization properly
4) Does not require a medium to propagate.	4) Requires a hypothetical ether medium.

[☞ Remaining VSAQs are in Page 96]

11 DUAL NATURE OF RADIATION & MATTER

(2 Marks)

153. What are cathode rays? [AP 17,18,23]

A: 1) **Cathode Rays are streams of electrons.**
2) These rays are emitted from cathode of a discharge tube, when the gas in it is subjected to low pressure (0.001 mm of Hg) and strong electric field.

---

154. What important fact did Millikan's oil drop experiment establish?

A: 1) Millikan's experiment established the fact that electric charge is quantised.
2) Millikan found the value of charge of electron is $e = 1.602 \times 10^{-19}$ C.

---

155. What is photoelectric effect? [AP 17,18,20]

A: **Photoelectric Effect** is the emission of electrons from a metal surface when it is illuminated by electro magnetic radiation of sufficient frequency. The emitted electrons are called **photoelectrons.**

---

156. Define work function. Give an expression for it.

A: 1) **Work Function:** The minimum energy required by an electron to escape from the surface of a metal is called work function (ϕ_0) of the metal.
2) **Expression:** $\phi_0 = hv_0$ Here h is planck's constant and v_0 is threshold frequency.

---

157. Give examples of photosensitive substances. Why are they called so? [AP 16]

A: 1) Alkali metals like lithium, sodium, potassium, cesium and rubidium are photosensitive substances.
2) Because, these alkali metals emit electrons when they are illuminated by light.

---

158. Write down Einstein's photoelectric equation. Explain the terms in it? [BMP]

A: **Einstein's Photoelectric Equation :** $K_{max} = hv - \phi_0$
Here K_{max} is maximum kinetic energy of the photoelectron, h is Planck's constant, v is the frequency of incident light and ϕ_0 is the work function of the metal.

---

159. Write down de-Broglie's relation and explain the terms therein. [AP 16,18,22]

A: **De Broglie's relation :** Wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$
Here h = Planck's constant and p = mv is the momentum of the particle

---

160. Define 1 electron volt.

A: An **electron volt** is a tiny unit of energy, defined as the kinetic energy gained by a single electron when it accelerates through a potential difference of 1 volt in vaccum. $1\,eV = 1.6 \times 10^{-19}$ J

---

161. Which physical quantities are conserved in photon-particle collisions in photoelectric effect?

A: In a photon- particle collision, the **total energy** and **total momentum** are conserved.

12 ATOMS

2 Marks

162. Define impact parameter. [BMP]

A: The **impact parameter (b)** is defined as the perpendicular distance of the initial velocity vector of the α– particle from the centre of the nucleus.

---

163. Define distance of closest approach.

A: **Distance of closest approach** is the minimum distance to which a fast moving α particle can approach the nucleus when it collides head-on before its KE is completely converted into electrostatic PE.

---

164. What is the difference between α - particle and helium atom?

A: 1) Helium atom is electrically neutral whereas alpha particle is positively charged.
2) Helium atom consists of 2 protons, 2 neutrons and 2 electrons whereas alpha particle is a double ionized helium ion (He^{2+}) consisting of 2 protons and 2 neutrons but with zero electrons.

---

165. Why did most of the α - particles pass through the gold foil undeflected in Rutherford's scattering?

A: Most of the α particles passed straight through the gold foil undeflected.
Because most of the space inside an atom is empty.
Thus nucleus occupies only a small fraction of total volume of atom.

---

166. What is the angular momentum of an electron in the second orbit of Bohr's model of hydrogen atom?

A: From Bohr's model of hydrogen atom, the angular momentum of electron in n^{th} orbit is $L = \frac{nh}{2\pi}$
For n = 2, $L = \frac{2h}{2\pi} = \frac{h}{\pi}$

---

167. Express the magnitude of electrostatic force of repulsion between the α - particle and the positively charged nucleus?

A: From Coulomb's law, Force $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{r^2}$
Here r is the distance between the alpha particle and the nucleus, (2e) is the charge of the alpha particle, and (Ze) is the charge of the nucleus.

[☞ Remaining VSAQs are in Page 97]

13 NUCLEI

2 Marks

168. What are isotopes and isobars? [BMP]

A: 1) **Isotopes** are the nuclei having same atomic number (Z), but different mass numbers (A)

Ex: $_1H^1$, $_1H^2$, $_1H^3$

2) **Isobars** are the nuclei having same mass number (A), but different atomic numbers (Z)

Ex: $_6C^{14}$, $_7N^{14}$

---

169. What are isotones and isomers?

A: 1) **Isotones** are the nuclei having the same neutron number (N), but different atomic numbers (Z).

Ex: $_{20}Ca^{40}$, $_{19}K^{39}$. Here number of neutrons in each = 20

2) **Isomers** are the nuclei having same atomic number (Z) and same mass number (A) but different nuclear properties such as radioactive decay and magnetic moments.

Ex: $^{80}_{35}Br$ (metastable), $^{80}_{35}Br$ (ground state)

---

170. Why are nuclear fusion reactions called thermo nuclear reactions?

A: 1) **Nuclear fusion** reactions take place at extremely high temperature (of the order 10^7K) in order to overcome strong electrostatic repulsions between positively charged nuclei.

2) Hence, the nuclear fusion reactions are called **thermo nuclear reactions.**

---

171. Compare the radii of the nuclei of mass numbers 27 and 64.

Sol: Given $A_1 = 27, A_2 = 64, \frac{R_1}{R_2} = ?$

Nuclear radius $R = R_0 A^{1/3} \Rightarrow R \propto A^{1/3} \Rightarrow \frac{R_1}{R_2} = \frac{\sqrt[3]{A_1}}{\sqrt[3]{A_2}} = \frac{\sqrt[3]{27}}{\sqrt[3]{64}} = \frac{3}{4}$

$\therefore R_1 : R_2 = 3 : 4$

---

172. What is the significance of a mass defect in a nucleus?

A: 1) The mass defect represents the **stability of the nucleus.**

2) A large **mass defect** generally means a larger binding energy, which makes the nucleus more stable. It proves that mass and energy are interchangeable according to $E = mc^2$.

[☞ Remaining VSAQs are in Page 98]

14 SEMI CONDUCTORS

2 Marks

173. What are intrinsic and extrinsic semiconductors ?

A: 1) **Intrinsic Semiconductor:**

Pure semiconductor like silicon is called intrinsic semiconductor.

In an intrinsic semiconductor, number of electrons(n_e) = number of holes(n_h).

2) **Extrinsic Semiconductor:** A semiconductor doped with trivalent or pentavalent impurity is called extrinsic semiconductor. Extrinsic semiconductors are of two types.

(i) n-type semiconductor and (ii) p-type semiconductor.

---

174. What is an n-type semiconductor?

What are the majority and minority charge carriers in it?

A: 1) **n-type Semiconductor:**

Silicon or Germanium doped with pentavalent impurity like Arsenic is called n-type semiconductor. The pentavalent atoms add extra electrons and make it n-type semiconductor.

2) In n-type semiconductor, **majority charge carriers** are **electrons** and **minority charge carriers** are **holes.**

---

175. What is a p-type semiconductor ?

What are the majority and minority charge carriers in it?

A: 1) **p-type Semiconductor:**

Silicon or Germanium doped with trivalent impurity like Indium is called p-type semiconductor. The trivalent atoms add extra holes and make it p-type semiconductor.

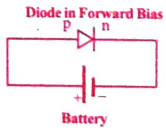
2) In p-type semiconductor, **majority** charge carriers are **holes** and **minority** charge carriers are **electrons.**

---

176. How is a battery connected to a junction diode in (1) forward and (2) reverse bias? [BMP]

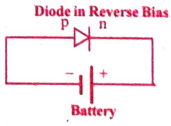
A: 1) **Forward bias:**

In forward bias, a battery is connected to the diode such that its positive terminal is towards p-end of diode.



2) **Reverse bias:**

In reverse bias, a battery is connected to the diode such that its positive terminal is towards n-end of diode.



177. What do you mean by donor impurity and acceptor impurity?

A: 1) **Donor impurities** are atoms with five valence electrons-like Phosphorus (Pentavalent) added to semiconductors to donate free electrons forming n-type materials.

2) **Acceptor impurities** are atoms with three valence electrons- like Boron (trivalent) added to semiconductors to create vacancies (holes) forming p-type materials.

---

178. What is reverse saturation current of p-n junction diode?

A: **Reverse saturation current** is the small amount of current that flows through a p-n junction diode when it is reverse biased. It is caused by the flow of minority carriers.

---

179. Define depletion layer and what is the breakdown voltage of p-n junction diode.

A: 1) **Depletion layer:** It is the narrow region on either side of the p-n junction which is free from mobile charge carriers.

2) **Breakdown voltage:** It is the voltage in a reverse biased p-n junction at which the current suddenly rises to a large values.

---

180. What happens to the width of the depletion layer in a p-n junction diode when it is i) forward-biased and ii) reverse biased?

A: i) In **forward bias**, the width of the depletion layer of a diode **decreases**

ii) In **reverse bias**, the width of the depletion layer of a diode **increases**.

---

181. What is rectification? Which device is used as rectifier?

A: 1) **Rectification** is the process of converting an alternating current in to direct current.

2) The device used for this rectification process is called a **rectifier.**



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[☞ Remaining VSAQs are in Page 98]